

Factor-Augmented VARs with Noisy Factor Proxies

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ABSTRACT

In factor-augmented vector autoregression (FAVAR) models, some of the factors are treated as observable while the remaining factors are latent and need to be estimated from a large cross-section of time series. Given that economic concepts such as inflation or output can often be proxied by different variables and that macroeconomic time series are commonly subject to substantial data revisions, the assumption that some factors are perfectly observable appears unnecessarily strong. In this paper we relax the assumption and treat observable factor proxies as noisy measures of true underlying factors. We show that when there are more observable proxies than true underlying factors, the standard FAVAR models reduce to a dynamic factor model (DFM) with a rank constraint. We propose an iterative estimation procedure that alternates between principal components estimation and solving a reduced-rank regression model. We further discuss a modification of the method with group-lasso sparsity constraints to incorporate regularization and variable selection at the same time. We use Monte Carlo simulations to demonstrate the effectiveness of the proposed method for factor estimation and forecasting in a DFM with weak factors and its usefulness for estimating structural impulse responses to oil supply and demand shocks using a FAVAR model.

Keywords: factor-augmented vector autoregressions, noisy factor proxies, principal components, reduced-rank regression, sparse regression

JEL Classification: C32, C38, C55, E17.

1 Introduction

Since [Bernanke, Boivin and Elias \(2005\)](#), factor-augmented vector autoregressions (FAVARs) have become a workhorse model in empirical macro-financial economics. A key assumption made in these models is that some factors are perfectly measured by observable time series while the other factors need to be extracted from a large panel of time series using principal components. However, the assumption of some factors being perfectly observed appears unnecessarily strong. First, many economic time series are subject to substantial data revisions. Second, economic concepts such as inflation or economic activity are rarely proxied by only one variable and are thus likely observed with measurement error.

In this paper we treat observable factor proxies as noisy measures of the true underlying factors. We show that when there are more observable factor proxies than true factors, the standard FAVAR models reduce to a dynamic factor model (DFM) with a rank constraint. We hence propose to estimate the model by an iterative procedure which alternates between principal components estimation and solving a reduced-rank regression model ([Anderson, 1951](#)). We further extend the method by adding a group-lasso type penalty ([Yuan and Lin, 2006](#); [Chen and Huang, 2012](#); [Qian, Tanigawa, Li, Tibshirani, Rivas and Hastie, 2022](#)) to leverage a sparsity assumption when a subset of factor proxies is believed to be particularly relevant to the prediction. Building on [Cheng, Liao and Schorfheide \(2016\)](#), we propose a cross-validation procedure to jointly determine the number of common factors along with the rank and penalty parameters.

We document the desirable properties of the proposed method via Monte Carlo simulations. First, we use a DFM design with strong and weak factors. The weak-factor cases we consider are motivated by previous Monte Carlo and empirical evidence suggesting that the choice of data used in factor estimation strongly affects the estimated factor space (see e.g., [Boivin and Ng, 2006](#); [Bai and Ng, 2008](#)): if the factor is related to only a small share of all observed series or explains only a small fraction of their overall variation, the factor is weak in the cross-section used for the analysis and the ability of standard principal components to estimate the factor space can deteriorate substantially. In principle, the standard FAVAR framework is well-suited to tackle the empirical issues related to weak factors, as it allows for inclusion of observable series as proxies for factors which are not well captured by the cross-section. However, this advantage vanishes when the observable factors are measured with error. We document that our approach of combining principal components with sparse reduced-rank regressions (RRR henceforth) considerably improves factor estimation and produces superior forecasts.

As a second illustration of the strength of our approach in the presence of noisy factor proxies, we use a data-based simulation design inspired by [Stock and Watson \(2016\)](#) to identify oil supply and demand shocks in the spirit of [Kilian \(2009\)](#). Specifically, we extend their analysis by including an alternative measure of global economic activity due to [Baumeister and Hamilton \(2019\)](#). We treat both global activity measures as noisy proxies for a single underlying factor and show that our proposed estimator substantially reduces the bias induced by using noisy observable series in estimating structural impulse response functions (SIRFs).

Our estimation approach builds on several literatures. The first is the large body of work related to FAVAR estimation via iterative principal components. This method was first developed by [Stock and Watson \(2005\)](#) and [Boivin, Giannoni and Mihov \(2009\)](#). We extend this estimator to include an additional reduced-rank regression step which closely follows [Izenman \(1975\)](#) and [Velu and Reinsel \(1998\)](#). We further introduce a group-lasso penalty for variable selection and regularization using the algorithm proposed by [Chen and Huang \(2012\)](#) and further extended by [Qian et al. \(2022\)](#).

A second related literature is the one on DFMs in the presence of weak factors. The standard principal components estimator assumes that the series used in the analysis have been chosen so that they contain useful information for factor estimation. However, this does not necessarily hold in large datasets. One approach to handle this issue is to model factors as being related to subsets of the data in the context of hierarchical DFMs (see e.g., [Moench, Ng and Potter, 2013](#)). Another approach is to use screening methods to prune the data as in [Bai and Ng \(2008\)](#), a method developed further by [Giglio, Xiu and Zhang \(2025\)](#) in an asset pricing context. Alternatively, one can use sparsity-inducing methods to select a subset of variables, for example by adding a lasso type penalty (e.g., see [Uematsu and Yamagata 2023a,b](#)). All these methods use either *a-priori* knowledge or data-driven approaches to screen out useless variables. In this paper, we instead use FAVARs to incorporate information from observable noisy proxies into the space of factors, and evaluate the properties of our estimator when some factors are weak. Using Monte Carlo simulations, we show that our reduced-rank principal component approach outperforms other methods including the one by [Uematsu and Yamagata \(2023a\)](#) in factor estimation and forecasting in the presence of weak factors.¹

Finally, a related recent paper is [Li, Plagborg-Møller and Wolf \(2024\)](#) who explore the properties of various estimators for SIRFs, including local projections and structural VARs. The authors use the DFM from [Stock and Watson \(2016\)](#) and [Lazarus, Lewis, Stock and Watson \(2018\)](#) as their encompassing model, from which they form thousands of lower dimensional DGPs. As in [Li et al. \(2024\)](#), we parameterize our second Monte Carlo design following [Stock and Watson \(2016\)](#), with the main difference being that we focus on FAVAR estimators of SIRFs in the presence of noisy observable factor proxies.

The paper is organized as follows. Section 2 describes the methodology and details the estimation procedure. We discuss the model selection approach and the normalization of the factors for SIRF analysis in Section 3. Section 4 lays out the DFM design and explores the finite-sample properties of the proposed method in terms of factor estimation and constructing forecasts. Section 5 presents a data-driven Monte Carlo analysis to study the bias induced from the use of noisy observable proxies in estimating SIRFs. Section 6 concludes.

¹For other important work studying strong and weak factors and statistical methods for measuring factor strength, see e.g., [Chudik, Pesaran and Tosetti \(2011\)](#), [Onatski \(2010, 2012\)](#), [Kapetanios and Marcellino \(2010\)](#), [Bailey, Kapetanios and Pesaran \(2021\)](#), and [Uematsu and Yamagata \(2023a\)](#).

2 Methodology

In this section we begin with a discussion of the standard FAVAR framework. Later on we will make important modifications to this framework and discuss the estimation procedure.

2.1 Framework

To fix ideas, consider a FAVAR with q observed factors $\tilde{F}_t$ measured without error by the variables Y_t , and r unobserved factors stacked in the vector F_t . These factors drive the common variation in a set of observed time series variables stacked in a $N \times 1$ vector X_t . Both, the observable variables X_t and the factor proxies Y_t are assumed to be second-order stationary and integrated of order zero. N is commonly assumed to be large. In particular, it may be greater than T , the number of time periods. It is also commonly assumed to be much greater than the total number of factors in the FAVAR model, i.e. $N \gg r + q$.

In typical applications, Y_t would e.g. include a policy rate, a measure of real economic activity, and a measure of inflation (e.g. [Bernanke et al., 2005](#)). Hence, the FAVAR model can be used to trace impulse responses to the observed variables Y in a data-rich environment, taking into account the joint dynamics of Y_t with a small number of factors F_t which span the information in a large number of additional variables X_t .

The FAVAR then takes the following form:

$$\begin{bmatrix} Y_t \\ X_t \end{bmatrix} = \begin{bmatrix} I_q & 0_{q \times r} \\ \Lambda_{X\tilde{F}} & \Lambda_{XF} \end{bmatrix} \begin{bmatrix} \tilde{F}_t \\ F_t \end{bmatrix} + \begin{bmatrix} 0 \\ e_{Xt} \end{bmatrix} \quad (1)$$

$$F_t^+ = \Phi(L) F_t^+ + \eta_t, \quad \text{where } F_t^+ = \begin{bmatrix} \tilde{F}_t \\ F_t \end{bmatrix} \quad (2)$$

where $\Lambda_{X\tilde{F}}$ and Λ_{XF} are the $N \times q$ and $N \times r$ loading matrices, $\Phi(L)$ is a lag polynomial of dimension $(r + q) \times (r + q)$, e_{Xt} is the $N \times 1$ vector of idiosyncratic errors and η_t is the $(r + q) \times 1$ vector of factor innovations. Equations (1) and (2) capture the main feature of FAVARs: the common dynamics of Y_t and X_t are driven by both $\tilde{F}_t$ and F_t , with $\tilde{F}_t$ treated as perfectly observed by Y_t .

In many applications, the assumption of observed factors which are measured without error appears unduly strong. For example, there are different variables which are proxies for economic concepts such as economic activity and inflation. Moreover, macroeconomic variables are often observed with a lag and feature substantial data revisions. As a consequence, in relevant applications it seems more appealing to assume that observable variables are proxies for the true underlying factors and are measured with error.

In what follows, we will therefore interpret Y_t as observed proxies which may be noisy measures of the true underlying factors. To be concrete, suppose we observe m proxy variables Y_t . In principle, m could be greater than the number of underlying factors $\tilde{F}_t$ ($m \geq q$). We thus assume that (X_t, Y_t)

admit the following factor model representation:

$$\begin{bmatrix} Y_t \\ X_t \end{bmatrix} = \begin{bmatrix} \Lambda_{Y\tilde{F}} & 0_{m \times r} \\ \Lambda_{X\tilde{F}} & \Lambda_{XF} \end{bmatrix} \begin{bmatrix} \tilde{F}_t \\ F_t \end{bmatrix} + \begin{bmatrix} e_{Yt}/\sqrt{T} \\ e_{Xt} \end{bmatrix}, \quad (3)$$

$$F_t^+ = \Phi(L) F_t^+ + \eta_t, \quad \text{where } F_t^+ = \begin{bmatrix} \tilde{F}_t \\ F_t \end{bmatrix} \quad (4)$$

where $\Lambda_{Y\tilde{F}}$ is $m \times q$ with full-column rank, and $e_{Yt}/\sqrt{T}$ is a $m \times 1$ vector of measurement errors with mean zero that is assumed to be uncorrelated with e_{Xt} .

Equations (3) and (4) make one important modification to (1) and (2): the variables Y_t observe the true factors with measurement errors that diminish at rate $T^{-1/2}$. In other words, the factors $\tilde{F}_t$ can be well-approximated by linear combinations of Y_t when T is large.² This permits rewriting equation (3) as

$$X_t = \Lambda_{X\tilde{F}} B' \left(Y_t - \frac{e_{Yt}}{\sqrt{T}} \right) + \Lambda_{XF} F_t + e_{Xt} = \Lambda_{X\tilde{F}} B' Y_t + \Lambda_{XF} F_t + u_{Xt}, \quad (5)$$

where $B = \Lambda_{Y\tilde{F}} \left(\Lambda'_{Y\tilde{F}} \Lambda_{Y\tilde{F}} \right)^{-1}$ is of dimension $m \times q$, and the error term vector is $u_{Xt} = e_{Xt} - \Lambda_{X\tilde{F}} B' \left(e_{Yt}/\sqrt{T} \right)$. Thus, equation (5) obtains with one restriction: the $N \times m$ coefficient matrix Λ_{XY} has rank q , and its $N \cdot m$ elements are expressed in terms of the $N \cdot q + m \cdot q$ elements of $\Lambda_{Y\tilde{F}}$ and B' . The main advantage of this representation is that the factors can be estimated using an iterative method that alternates between estimating B via reduced-rank regression for a given F_t and estimating F_t for a given estimate of B , see Section 2.2 below.

Note that our assumption about the errors e_{Yt} diminishing as $T \rightarrow \infty$ implies that the observables Y are "good" proxies of the true underlying factors $\tilde{F}$. In applied work, whether good proxies are used is ultimately an empirical question. We address this question in two ways. First, in Section 2.3, we discuss an extension of the estimation method by adopting a group lasso penalty on the rows of B to impose a sparsity assumption when a subset of factor proxies is believed to be relevant for the FAVAR estimation. Second, in Section 3.1 we introduce a model selection procedure via cross-validation that jointly determines the number of the unobserved factors r , the number of factor proxy combinations q , as well as the penalty parameter in the case with additional sparsity constraints. In Section 4, we show in a Monte Carlo analysis that both approaches perform very well specifically in the presence of weak noisy factors.

²This assumption follows the one in [Parker and Sul \(2016\)](#) for the case of approximate factor proxies. Moreover, [Schorfheide \(2005\)](#) and [Li et al. \(2024\)](#) assume drifting DGPs in the context of forecasting and analyzing impulse responses where the sample is not generated from a VAR(p), but the degree of misspecification vanishes at rate $T^{-1/2}$.

2.2 Estimation

If the factors $\tilde{F}_t$ are measured without error as in the standard FAVAR framework, the model can be estimated via the iterative principal component estimator proposed by [Stock and Watson \(2005\)](#) and [Boivin, Giannoni and Mihov \(2009\)](#). However, under the rank constraint, equation (5) cannot be estimated this way. Instead, we propose a method that combines iterative principal component estimation with an additional reduced-rank regression step ([Izenman, 1975](#); [Velu and Reinsel, 1998](#)).

Specifically, denoting the matrix $[A_1, \dots, A_T]'$ by A for any time series of vectors $\{A_t\}_{t=1}^T$, the factors can be estimated as the principal components of the filtered observations $(X - Y\hat{B}\hat{\Lambda}'_{X\hat{F}})$ given the parameter estimates $\hat{B}$ and $\hat{\Lambda}$. In turn, given the factor estimates $\hat{F}$, the coefficient matrices $\hat{B}$ and $\hat{\Lambda}$ can be estimated via a reduced-rank regression of $M_{\hat{F}}X$ on $M_{\hat{F}}Y$ where $M_{\hat{F}} = I_T - \hat{F}(\hat{F}'\hat{F})^{-1}\hat{F}'$. Specifically, the estimators of F , B , $\Lambda_{X\hat{F}}$ and Λ_{XF} jointly solve the minimization problem

$$\underset{F, B, \Lambda_{X\hat{F}}, \Lambda_{XF}}{\text{minimize}} \|X - YB\Lambda'_{X\hat{F}} - F\Lambda'_{XF}\|^2, \quad (6)$$

subject to the normalization constraints

$$\Lambda'_{X\hat{F}}\Lambda_{X\hat{F}} = I_q, \quad F'F = I_r, \quad \Lambda'_{XF}\Lambda_{XF} \text{ diagonal} \quad (7)$$

where $\|\cdot\|$ denotes the Frobenius norm. This minimization can be done iteratively. Given a preliminary estimator of F , which is computed as the first r principal components of X , the estimators for $\Lambda_{X\hat{F}}$, B and Λ_{XF} are given by:

$$\begin{aligned} \hat{\Lambda}_{X\hat{F}} &= V, \quad \hat{B} = (Y'M_{\hat{F}}Y)^{-1}Y'M_{\hat{F}}XV, \\ \text{and } \hat{\Lambda}_{XF} &= (X - Y\hat{B}\hat{\Lambda}'_{X\hat{F}})' \hat{F} (\hat{F}'\hat{F})^{-1} = (X - Y\hat{B}\hat{\Lambda}'_{X\hat{F}})' \hat{F}, \end{aligned}$$

where $V = [V_1, \dots, V_q]$ and V_j is the j th eigenvector of $X'M_{\hat{F}}Y(Y'M_{\hat{F}}Y)^{-1}Y'M_{\hat{F}}X$.³ Given estimates of B and $\Lambda_{X\hat{F}}$, we estimate F as the first r principal components of $(X - Y\hat{B}\hat{\Lambda}'_{X\hat{F}})$. Each step of this procedure reduces the sum of squares in equation (6) and the steps can be repeated until convergence.

The above-mentioned estimation procedure is justified by a large body of distribution theory concerning the behavior of principal components estimator in large panels and of reduced-rank regression estimators in multivariate linear models. [Stock and Watson \(2002\)](#) establish consistency of the principal components estimator for the factor space under conditions consistent with the approximate factor model of [Chamberlain and Rothschild \(1983\)](#). [Bai \(2003\)](#) further derives the asymptotic distribution of the estimated factor loadings, showing asymptotic normality under standard approximate factor model assumptions when $N \rightarrow \infty$, $T \rightarrow \infty$ and $\sqrt{T}/N \rightarrow 0$. [Bai and Ng \(2006\)](#) analyze the distribution of coefficients in factor-augmented regressions in which X_{it} is regressed on estimated factors $\hat{F}_t$ and a fixed number of additional observed stationary regressors, and show that standard $\sqrt{T}$ inference

³See [Velu and Reinsel \(1998\)](#) for an extensive discussion on RRR models and their estimation.

remains valid under the same rate conditions. Together, these results provide a formal justification for standard inference in FAVAR models: when both the cross-sectional and time dimensions are large, factors estimated by principal components may be treated as if they were observed, in the sense that factor estimation error does not affect first-order asymptotic inference.

In addition, our estimation approach uses a reduced-rank regression step to include information from observed noisy factor proxies in the factor estimation. This implies that variation associated with the factors will be captured by $\hat{F}^+ = [\tilde{Y}\hat{B}, \hat{F}]$, where $\tilde{Y} = M_{\hat{F}}Y$ denotes the residuals obtained from projecting Y onto $\hat{F}$. Note that this is equivalent to obtaining factor estimates applying the principal components estimator to $P_{\tilde{Y}}\tilde{X}$, where $\tilde{X} = M_{\hat{F}}X$ denotes the residuals of regressing X onto $\hat{F}$ and $P_{\tilde{Y}} = \tilde{Y}(\tilde{Y}'\tilde{Y})^{-1}\tilde{Y}'$ are the projection matrix onto $\tilde{Y}$. In other words, our reduced-rank regression estimator is equivalent to extracting principal components from a projection of $\tilde{X}$ onto $\tilde{Y}$, where $\hat{\Lambda}_{X\tilde{F}}$ is the matrix of the q leading eigenvectors of $\tilde{X}'P_{\tilde{Y}}\tilde{X}$, and $\tilde{Y}\hat{B} = P_{\tilde{Y}}\tilde{X}\hat{\Lambda}_{X\tilde{F}}$.

The key advantage of this estimation procedure becomes clear when we distinguish between strong and weak factor models. In a strong factor model, where the factors provide a strong signal and feature exploding eigenvalues in the covariance matrix as T and N approach infinity, our estimator is equivalent to the standard principal component estimator. However, in many applications, the factor model might include weak factors associated with eigenvalues that are large compared to the idiosyncratic components, but do not explode as the panel dimensions increase. In these cases, the standard principal components estimator does not capture the true factor space (see, e.g. [Kapetanios and Marcellino, 2010](#); [Onatski, 2012](#); [Uematsu and Yamagata, 2023a](#)). Our estimation approach is designed to use additional information from noisy observable factor proxies and as such is preferable in such environments. Monte Carlo evidence presented in Section 4 shows that this approach works well in comparison to the method of [Uematsu and Yamagata \(2023a\)](#) which is designed to capture weak factors by imposing sparsity on the factor loadings.

2.3 Modification with sparsity constraints

We now introduce an extension of our approach where a subset of factor proxies are believed to be particularly relevant for estimation of the factors and FAVAR parameters. This can be achieved by assuming sparsity in the coefficient matrix B . To find a sparse estimate of the coefficients, we propose to add a group lasso penalty to the objective function (6) and solve the following penalized rank-constrained optimization problem

$$\underset{F, B, \Lambda_{X\tilde{F}}, \Lambda_{XF}}{\text{minimize}} \quad \frac{1}{2} \|X - YB\Lambda'_{X\tilde{F}} - F\Lambda'_{XF}\|^2 + \lambda \sum_{j=1}^m \|B_{j\bullet}\| \quad (8)$$

subject to the same normalization (7) as above, where $B_{j\bullet}$ is the j th row of B , and $\lambda \geq 0$ is a penalty parameter. The penalty term enforces that if the fitted model has $\|B_{j\bullet}\| = 0$, i.e. all the elements in row $B_{j\bullet}$ equal 0, otherwise all the elements in $B_{j\bullet}$ will be nonzero.

This optimization problem can be solved using a similar iterative procedure as described in Section 2.2, except for one difference: we replace the RRR step with the alternating minimization method proposed by Chen and Huang (2012) and further extended by Qian et al. (2022). For a given estimate of factors $\hat{F}$ and λ value, the estimate of the parameters B , $\Lambda_{X\hat{F}}$ and Λ_{XF} are obtained by iterating two steps. Specifically, for the k th iteration:

1. Given an estimate $\hat{B}^{(k)}$ and $\hat{\Lambda}_{XF}^{(k)}$, we solve the orthogonal Procrustes problem

$$\underset{\Lambda_{X\hat{F}}}{\text{minimize}} ||X - Y\hat{B}^{(k)}\Lambda'_{X\hat{F}} - \hat{F}\hat{\Lambda}_{XF}^{(k)'}||^2$$

subject to $\Lambda'_{X\hat{F}}\Lambda_{X\hat{F}} = I_q$. Let $(X - \hat{F}\hat{\Lambda}_{XF}^{(k)'})'YB^{(k)} = UDV'$ be the Singular Value Decomposition with U , D and V having dimensions $N \times q$, $q \times q$ and $q \times q$, respectively. The solution is given as $\hat{\Lambda}_{X\hat{F}}^{(k+1)} = UV'$.

2. Given $\hat{\Lambda}_{X\hat{F}}^{(k+1)}$, we solve the group Lasso problem

$$B^{(k+1)} = \underset{B}{\text{argmin}} \frac{1}{2} ||M_{\hat{F}}X\hat{\Lambda}_{X\hat{F}}^{(k+1)} - M_{\hat{F}}YB||^2 + \lambda \sum_{j=1}^m ||B_{j\bullet}||,$$

where $M_{\hat{F}} = I_T - \hat{F}(\hat{F}'\hat{F})^{-1}\hat{F}'$, and $\hat{\Lambda}_{XF}^{(k+1)} = (X - YB^{(k+1)}\hat{\Lambda}_{X\hat{F}}^{(k+1)'})'\hat{F}$.⁴

Starting values are obtained from the standard RRR estimation we have described in the previous section, and the procedure is iterated until convergence.

3 Practical guidance for implementation

We now discuss our approach to model selection and normalization of the factors in structural FAVARs for impulse response analysis.

3.1 Model selection

In Section 2.2, we laid out the estimation procedure for a given number of latent factors r , the rank q associated with the set of noisy factor proxies, and the regularization parameter λ in the group Lasso approach. We now discuss how to determine these parameters in practice.

Our approach builds on the cross-validation procedure proposed by Cheng, Liao and Schorfheide (2016). Using Monte Carlo analysis, these authors show that the procedure works well compared to the methods proposed by Bai and Ng (2002), Onatski (2010) and Ahn and Horenstein (2013) in standard factor models with common factors estimated via principal components. What distinguishes

⁴We solve the group-lasso problem by `glmnet` (Friedman, Hastie and Tibshirani 2010) with the `mgaussian` family.

our setup is the existence of additional observable noisy factor proxies. We therefore adapt the cross-validation procedure of [Cheng, Liao and Schorfheide \(2016\)](#) to jointly determine r and q . Moreover, in the case of sparsity constraints on B , we also let the procedure determine λ jointly with r and q .

Let $0 \leq r \leq r_{max}$ and $0 \leq q \leq q_{max}$. Hence, the smallest model that can be chosen does not include any common factors or noisy factor proxies Y . For a given r , let there be a decreasing sequence of values $\lambda_{max} = \lambda_1 > \lambda_2 > \dots > \lambda_L \geq 0$, where λ_{max} is defined such that it gives rise to the trivial solution of B where all the elements in B equal zero and the remaining λ_i are selected from an equally spaced array on the log scale. Following [Qian et al. \(2022\)](#), we set $\lambda_{max} = \max_{1 \leq j \leq m} \|Y_j' (X - \hat{F} \hat{\Lambda}_{XF}')\|$, where Y_j' is the $T \times 1$ vector of observations for the j th factor proxy, $\hat{F}$ is the matrix of eigenvectors corresponding to the r largest eigenvalues of XX' , and $\hat{\Lambda}_{XF}$ is the corresponding estimate of the factor loading matrix. In other words, the value λ_{max} corresponds to the case where the noisy factor proxies Y carry no additional information over and above the factors $\hat{F}$.

A graphical illustration of the cross-validation procedure is given in [Figure 1](#). We first partition the matrix X along the cross-sectional dimension, creating K disjoint subsamples $X_{I_1}, X_{I_2}, \dots, X_{I_K}$ with $I_1 \cup I_2 \cup \dots \cup I_K = \{1, 2, \dots, N\}$. Given a particular combination (r, q) and the corresponding λ values, we estimate the factors $\hat{F}^{+(K)} = [Y \hat{B}^{(K)}, \hat{F}^{(K)}]$ using the variables in cross-sectional units in $(K - 1)$ folds. We then partition the hold-out sample X_{I_K} along the time dimension into regression and prediction samples. Using the factor estimates from the $(K - 1)$ folds as regressors, we employ a rolling pseudo-out-of-sample forecasting scheme where we estimate the factor loadings based on the regression sample via OLS. We then use the factors and estimated loadings to generate pseudo-out-of-sample forecasts for the prediction sample. We repeat these steps by using each of the cross-sectional folds as the left-out set. The parameters are then selected by minimizing the sum of squared cross-validation prediction errors among the possible choices of (r, q) and the corresponding λ values. In [Section 4](#), we study the performance of the model in conjunction with the cross-validation method in a Monte Carlo analysis.

3.2 Impulse responses and named factor normalization

From the model described above, it is clear that only the space spanned by F_t and $B'Y_t$ is identified, whereas the F_t and $B'Y_t$ themselves are not uniquely pinned down. Since the factors are unobserved, one commonly imposes normalization [\(7\)](#) which represents a mathematically convenient set of restrictions which solves the identification issue for typical applications such as forecasting. However, when a FAVAR model is used to identify structural shocks, the factors need to be assigned an economic interpretation. This requires a different set of identification restrictions. In what follows, we achieve this by adopting the “named-factor” normalization of [Stock and Watson \(2016\)](#). In this approach, each factor is aligned with a specific naming variable.⁵

Let

⁵[Bai and Ng \(2013\)](#) denote this the PC3 normalization.

$$X_t^+ = \begin{bmatrix} Y_t \\ X_t \end{bmatrix}, \quad \hat{F}_t^+ = \begin{bmatrix} \hat{B}'Y_t \\ \hat{F}_t \end{bmatrix}, \quad \text{and } \hat{\Lambda} \text{ be the estimate of the loadings of } X_t^+ \text{ on } \hat{F}_t^+, \quad (9)$$

where $\hat{F}_t$ and $\hat{B}$ are the estimators produced via the procedure previously described. Moreover, order X_t^+ so that the first $(r+q)$ variables are the naming variables (including some of the elements of Y_t), and let $\hat{\Lambda}_{1:q+r}$ denote the $(r+q) \times (r+q)$ matrix of the loadings on the first $(r+q)$ variables. The named-factor normalization is

$$\tilde{\Lambda} = \begin{bmatrix} I_{q+r} \\ \tilde{\Lambda}_\bullet \end{bmatrix} = \begin{bmatrix} I_{q+r} \\ \hat{\Lambda}_\bullet \hat{\Lambda}_{1:q+r}^{-1} \end{bmatrix}, \quad \text{and} \quad \tilde{F}_t^+ = \hat{\Lambda}_{1:q+r} \hat{F}_t^+. \quad (10)$$

The above named-factor normalization aligns factors with naming variables and sets the magnitude of the loadings so that a unitary innovation in factor i raises the i th naming variable by one unit on impact. As an example, suppose one of the naming variables is [Kilian \(2009\)](#)'s global activity measure. Then, the named-factor normalization associates the corresponding factor with the common component of [Kilian \(2009\)](#)'s measure.

4 Monte Carlo Analysis I: Factor Estimation and Forecasting

In this section, we discuss the finite-sample properties of the proposed estimator via simulations. The data are generated using a range of DFM specifications that feature serially correlated factors, and error terms that are serially and cross-sectionally correlated. Moreover, we study how the presence of weak factors affects the performance of the estimator for estimation of the factor space and for forecasting. We refer to weak factors as factors for which only a small proportion of the variables feature nonzero loadings.

4.1 Experimental design

Our experimental design is based on the following model

$$X_{it} = \lambda_{i1}F_{1t} + \lambda_{i2}F_{2t} + u_{it}, \quad (11)$$

$$\begin{bmatrix} F_{1t} \\ F_{2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & 0 \\ 0 & \phi_{22} \end{bmatrix} \begin{bmatrix} F_{1t-1} \\ F_{2t-1} \end{bmatrix} + \begin{bmatrix} \sqrt{\theta} & 0 \\ 0 & \sqrt{(1-\phi_{22}^2)} \end{bmatrix} \begin{bmatrix} \eta_{1t} \\ \eta_{2t} \end{bmatrix}, \quad (12)$$

$$(1 - aL)u_{it} = \sqrt{\nu}e_{it}, \quad \text{and} \quad (13)$$

$$e_{it} = (1 + b^2)\epsilon_{it} + b\epsilon_{i-1t} + b\epsilon_{i+1t}, \quad (14)$$

where all factor loadings λ_{ik} and disturbances η_{kt} and ϵ_{it} are iid $N(0, 1)$. This model is a static-form of a DFM with two factors following a VAR(1). We set $\phi_{11} = \phi_{22} = 0.5$. The first factor is assumed to be a common factor, i.e. there are no restrictions on the loadings. We consider two different cases for

the second factor. In the first, we set 50% and in the second, 90% of the loadings equal to zero. The parameter θ is chosen so that $\text{var}(F_{2t}) = \text{var}(F_{1t})$. We also entertain a case where we do not impose any restrictions on factor loadings but set the parameter θ such that $\text{var}(F_{2t}) = 0.1 \times \text{var}(F_{1t})$. The parameters a and b govern serial and cross-sectional correlation in the errors and are assumed to take values equal to 0.5 and 1, respectively. The parameter ν is chosen to ensure that the variables X_{it} have an average $R^2 = 30\%$ on the factors. This specification for the error terms builds on [Stock and Watson \(2002\)](#). Furthermore, the choice of average R^2 is consistent with common factor analyses of macroeconomic data ([Stock and Watson, 2016](#); [McCracken and Ng, 2016](#)).

We use noisy proxies for the second factor. Specifically, we set $Y_t = F_{2t} + \sigma\psi_t$ where ψ_t is iid $N(0, 1)$. We entertain five cases. In the first case we assume one proxy with an R^2 of 80% in the regression of Y_t on F_{2t} . We then consider a second case by adding four additional proxies with an R^2 of 70%, 60%, 50% and 40%, respectively. The third and fourth case respectively add five and ten additional proxies each having an R^2 of 30%. Modeling the proxies in this way allows us to consider cases where we start with the strongest proxy, but then add more and more proxies which are ever noisier measures of the true underlying factor. Finally, we also consider a fifth case with 20 proxies, where each proxy is generated to have an R^2 of 30%. We use datasets with dimensions $N = T = 100$ and for each set of parameters we conduct 1000 repetitions.

4.2 Estimation

We examine five estimation approaches.

1. *PC*: We estimate the factors via principal components and determine r by the edge distribution (ED) method of [Onatski \(2010\)](#). [Uematsu and Yamagata \(2023a\)](#) stress the importance of using this approach for determining r in weak factor models and provide Monte Carlo evidence suggesting that the method works well relative to those proposed by [Bai and Ng \(2002\)](#) and [Ahn and Horenstein \(2013\)](#) in weak factor model designs.
2. *SOFAR*: We estimate the factors via the adaptive sparse orthogonal factor regression (SOFAR) estimator of [Uematsu and Yamagata \(2023a\)](#) and determine r by [Onatski \(2010\)](#)'s ED method. Following [Uematsu and Yamagata \(2023a\)](#), we also use BIC to determine the regularization parameter for the adaptive SOFAR estimation.
3. *RPC_(1,1)*: We solve the optimization problem (6), setting $r = 1$ and $q = 1$. This is a benchmark that always combines one principal component with one combination of factor proxies.
4. *RPC*: We solve the optimization problem (6), and determine r and q using the cross-validation procedure detailed in [Section 3.1](#).
5. *SRPC*: We solve the penalized optimization problem (8), and determine r , q and λ via the cross-validation procedure detailed in [Section 3.1](#). This is a modification of *RPC* with sparsity constraints.

4.3 Summary statistics

We use two summary statistics to assess the relative performance of the four estimation procedures. The first is [Stock and Watson \(2002\)](#)'s trace R^2 of the regression of $\hat{F}^+$ onto F :

$$R_{\hat{F}^+, F}^2 = \frac{\widehat{E} \|P_F \hat{F}^+\|^2}{\widehat{E} \|\hat{F}^+\|^2} = \frac{\widehat{E} \text{tr} \left(\hat{F}^{+'} P_F \hat{F}^+ \right)}{\widehat{E} \text{tr} \left(\hat{F}^{+'} \hat{F}^+ \right)}, \quad (15)$$

where $\widehat{E}$ denotes the average over all the Monte Carlo repetitions, and $P_F = F(F'F)^{-1}F'$. The second statistic measures the out-of-sample forecast performance for the cross-section of variables, which also follows [Stock and Watson \(2002\)](#). The out-of-sample forecast is $\hat{X}_{i,T+1|T} = \hat{\beta}_i' \hat{F}_T^+$ where the OLS estimate $\hat{\beta}_i$ is obtained by regressing X_{it+1} on $\hat{F}_t^+$ for $t = 1, \dots, T-1$. The measure of relative forecast performance is

$$\frac{S_{(S)RPC}^2}{S_{PC}^2} = \frac{\sum_{i=1}^N \widehat{E} \left(\hat{X}_{i,T+1|T}^{(S)RPC} - X_{i,T+1|T} \right)^2}{\sum_{i=1}^N \widehat{E} \left(\hat{X}_{i,T+1|T}^{PC} - X_{i,T+1|T} \right)^2}. \quad (16)$$

Hence, we use the standard PC estimator as the benchmark, with r determined via [Onatski \(2010\)](#)'s ED method. Thus, the statistic is less than one when the out-of-sample forecasting performance of a model combining principal components with combinations of factor proxies outperforms those using only principal components.

4.4 Results

Table 1 reports the results of the Monte Carlo simulations. Panel A presents the results for the case in which there are two common factors. Panel B shows the corresponding results for the case where F_{1t} is a common factor, and 50% of the loadings are set to zero for F_{2t} , with $\text{var}(F_{2t}) = \text{var}(F_{1t})$. Panel C.1 shows the corresponding results for the case where F_{1t} is a common factor, F_{2t} is a weak factor with 90% of its loadings set to zero and $\text{var}(F_{2t}) = \text{var}(F_{1t})$. Panel C.2 presents the corresponding results for the case where F_{1t} is a common factor, F_{2t} is a weak factor without any restriction imposed on its loadings, but now $\text{var}(F_{2t}) = 0.1 \times \text{var}(F_{1t})$. We also report the average number of factors r determined by [Onatski \(2010\)](#)'s ED method as well as the average r and the rank q of factor proxies determined via the cross-validation procedure described in Section 3.1 under both the classical reduced-rank regression setting and its modification with sparsity constraints.

Consider first the results for the case of two common factors (panel A). We see that almost all of the procedures perform well in estimating the factor space with values of $R_{\hat{F}^+, F}^2$ around 0.93 for most specifications. The exception is $\text{RPC}_{(1,1)}$. The $R_{\hat{F}^+, F}^2$ value for $\text{RPC}_{(1,1)}$ is 0.86 when using one proxy. However, its performance quickly improves when adding more, albeit noisier proxies of the second factors. For example, when using 20 proxies, the value of $R_{\hat{F}^+, F}^2$ for $\text{RPC}_{(1,1)}$ equals 0.94.

We also see that combining principal components with only one noisy factor proxy leads to a deterioration of the out-of-sample performance of the method. In this case, the average cross-validation

estimate of q , $\hat{q}_{CV}$, is close to zero, suggesting the model does not include observed proxies from factor estimation in this case. However, this finding reverses when more noisy factor proxies are included in the estimation. The relative forecast performance quickly improves and the estimated values of $\hat{q}_{CV}$ approaching one indicate that in most simulations the procedure includes one common noisy factor.

Consider now the case in which 50% of the loadings are zero for one of the factors. The results are summarized in Panel B. The values of $R^2_{\hat{F}^+, F}$ for PC around 0.87 shows that the standard PC approach performs slightly worse estimating the factor space in the presence of a mildly weak factor. The SOFAR method of [Uematsu and Yamagata \(2023a\)](#) implies slightly higher values of explained factor variation, illustrating the usefulness of their approach in the presence of weak factors. That said, the $R^2_{\hat{F}^+, F}$ values further increase for our RPC and SRPC method when the number of noisy factor proxies is greater than one. Even more strikingly, the forecasting power, summarized by the values of $\frac{S^2_{(S)RPC}}{S^2_{PC}}$, improves dramatically for both methods in this case. When 20 proxies are used, both approaches essentially halve the forecast error relative to the standard PC method.

Finally, let us consider the cases where the second factor is very weak, summarized in Panels C.1 & C.2. Panel C.1 shows that the performance of the standard PC method substantially deteriorates in the case where weakness of the second factor is introduced by sparse loadings. While the SOFAR method appears unable to substantially improve the factor space estimation, the RPC and SRPC methods fare somewhat better in this dimension. Moreover, they substantially improve the forecast performance, especially for intermediate numbers of noisy factor proxies.

The last panel of Table 1 introduces weakness in the second factor not via sparse loadings but through a substantially smaller factor variance. As before, the standard PC method performs poorly in this case, and while the SOFAR method improves the estimation of the factor space, its forecasting performance is not much better than that of the standard PC approach. Turning to the RPC and SRPC methods, we see that both substantially improve the estimation of the factor space even for small numbers of noisy factor proxies. Moreover, they dramatically improve the forecasting performance of the model.

In summary, the main insight we glean from these experiments is that the use of RPC and SRPC substantially improves factor estimation and forecast performance in the presence of weak factors. At the same time, when there is no weak factor, the methods do not come at an expense relative to the standard PC approach.

5 Monte Carlo Analysis II: Structural Impulse Response Functions

In this section, we use data-based simulations to illustrate the desirable properties of our proposed estimator for the identification of structural economic shocks using a FAVAR model in the presence of noisy observable factors. Specifically, we identify structural shocks driving factor innovations, with or without measurement error. Since we know the true structural impulse responses, we can assess the degree to which measurement error in the factors induces a bias in FAVAR-based estimates of

structural impulse responses.

For illustration, consider the following simple case with two proxies for a single unobserved factor. Suppose that the researcher observes two proxies of which one is measured without error while the other contains some measurement noise. The researcher does not know which one is the proxy without noise. Now estimate three FAVARs: (1) with the perfectly observed factor proxy; (2) with the noisy proxy; and (3) with a linear combination of the two noisy proxies estimated using our reduced-rank approach setting.

Our object of interest is the structural impulse response to a shock underlying the factor. We assess the bias induced by the noise in the proxy, and show how the results differ across the three FAVAR specifications.

5.1 Data

Our data are quarterly observations on 208 time series taken from [Stock and Watson \(2016\)](#). This dataset consists of time series covering the most important categories of U.S. economic activity, as well as variables representing the oil market and global economic activity. Each series was transformed to be approximately stationary by taking growth rates or first differences.

A small number of outliers were replaced by the median of the preceding five periods, and the long-run mean of each series was subtracted using a biweight low-pass filter with a bandwidth of 100 quarters. Note that the original [Stock and Watson \(2016\)](#) dataset consists of 207 series. We make two small changes to the data. First, we update [Kilian \(2009\)](#)'s global activity index to its more recent [Kilian \(2019\)](#) version. Second, we include the growth rate of IP for the OECD countries and the six largest Emerging Market economies from [Baumeister and Hamilton \(2019\)](#) as an additional proxy of global activity. We follow [Stock and Watson \(2016\)](#) and use the sample from 1985Q1 to 2014Q4.

5.2 Experimental design

Our Monte Carlo analysis is presented in the context of identifying oil and global demand shocks. We consider the structural impulse responses obtained by using the recursive identification scheme of [Kilian \(2009\)](#) which distinguishes between three shocks: an oil supply shock, a global demand shock, and an oil-specific demand shock. We use FAVAR DGPs and empirically calibrate them following [Stock and Watson \(2016\)](#)'s implementation of the [Kilian \(2009\)](#) identification. In their implementation, three of the factors are treated as observed: the real oil price (PPI-oil), global oil production and a measure of global demand. The remaining five factors are extracted from the large panel of U.S. macroeconomic time series. We consider two measures of global demand: Kilian's global activity index, and the growth rate of IP of the OECD + 6 Emerging Markets following [Baumeister and Hamilton \(2019\)](#).

Identification We follow [Kilian \(2009\)](#) and use Cholesky decompositions to separately identify an oil supply shock, a global demand shock, and an oil-specific demand shock. To achieve this, we order global oil production, global demand measure and the real price of oil as the first three variables in

the FAVAR and adopt a Cholesky factorization of the shocks. Moreover, we adopt the “unit-effect” normalization of [Stock and Watson \(2016\)](#), so that a unit increase in each shock increases the associated factor innovation by one unit.

FAVAR DGPs and Structural Impulse Response Functions We now turn to the description of the DGPs, which are constructed based on [Stock and Watson \(2016\)](#)’s implementation of the [Kilian \(2009\)](#) identification scheme. Consider the FAVAR of [Stock and Watson \(2016\)](#) with eight factors:

$$\begin{bmatrix} \text{Oil Production}_t \\ \text{Kilian's Global Activity}_t \\ \text{Oil Price}_t^{\text{PPI}} \\ X_{\bullet,t} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \\ & & & \Lambda_{\bullet} & & \end{bmatrix} \begin{bmatrix} \text{Oil Production}_t \\ F_t^{\text{Global Activity}} \\ \text{Oil Price}_t^{\text{PPI}} \\ F_{4:8,t} \end{bmatrix} + u_t, \quad (17)$$

$$\begin{bmatrix} \text{Oil Production}_t \\ F_t^{\text{Global Activity}} \\ \text{Oil Price}_t^{\text{PPI}} \\ F_{4:8,t} \end{bmatrix} = \Phi(L) \begin{bmatrix} \text{Oil Production}_{t-1} \\ F_{t-1}^{\text{Global Activity}} \\ \text{Oil Price}_{t-1}^{\text{PPI}} \\ F_{4:8,t-1} \end{bmatrix} + \eta_t, \quad \text{Cov}(\eta_t) = \Sigma_{\eta}, \quad \text{and} \quad (18)$$

$$u_{it} = a_i(L)u_{it-1} + e_{it} \quad i = 1, 2, \dots, 208, \quad (19)$$

where all of the disturbances e_{it} are iid $N(0, \sigma_i^2)$, all of the factor innovations η_t are iid $N(0, \Sigma_{\eta})$, $\Phi(L)$ is a VAR(4) lag polynomial, $a_i(L)$ is an AR(4) lag polynomial, and the unit elements on the diagonal of the upper 8×8 partition of the loading matrix impose the named-factor normalization so that $F_t^{\text{Global Activity}}$ (the global activity factor) is the common component of Kilian’s global activity index.

We consider three DGPs in the context of the FAVAR model summarized by equations (17)-(19). Although all three DGPs augment global oil production, a global activity measure and the real oil price with five unobserved factors, they differ in terms of the global activity measure included in the factor space. Specifically, in the first DGP, we equate $F_t^{\text{Global Activity}}$ to Kilian’s global activity index. This is exactly the model of [Stock and Watson \(2016\)](#). In the second DGP, we use the OECD + 6 IP growth from [Baumeister and Hamilton \(2019\)](#) instead of Kilian’s index as the global activity index. In the third DGP we treat both Kilian’s index and [Baumeister and Hamilton \(2019\)](#)’s OECD + 6 IP growth as noisy measures of global activity and use a linear combination of the two as the true global activity factor.⁶

The parameters underlying the first two DGPs are estimated by implementing the models following [Stock and Watson \(2016\)](#). The parameters underlying the third DGP are obtained by estimating the model with both global activity proxies via our RPC approach which iteratively combines principal components and reduced-rank estimation. The estimated models are then used to simulate data by independently drawing $e_{it} \sim N(0, \hat{\sigma}_i^2)$ and $\eta_t \sim N(0, \hat{\Sigma}_{\eta})$.

⁶The weights used in this linear combination are 1.64 for the Kilian index and 2.4 for the Baumeister-Hamilton index, corresponding to the estimated weights applying the RPC approach to the actual data.

We now define the structural impulse responses which are the main objective criterion in this Monte Carlo analysis. We formulate the responses in terms of the level of each series. This is accomplished by using cumulative impulse responses for those series that enter the model after first or second differencing. We assume that the factor innovations η_t are linear combinations of structural shocks:

$$\eta_t = H\epsilon_t. \quad (20)$$

The recursive identification of Kilian (2009) with the unit-effect normalization implies $H = \text{Chol}(\Sigma_\eta)\Sigma_\epsilon^{-1/2}$, where $\Sigma_\epsilon = \text{diag}\left(\left\{\text{Chol}(\Sigma_\eta)_{jj}^2, j = 1, \dots, 8\right\}\right)$ with Chol denoting the Cholesky factorization. Then, the structural impulse responses are

$$\text{SIRF}_{ij}(h) = \begin{cases} C_{ij,h} & \text{if } X_{it} \text{ is in levels} \\ \sum_{k=0}^h C_{ij,k} & \text{if } X_{it} \text{ is differenced} \end{cases}, \quad \text{and} \quad C_h = \Lambda B_h H, \quad (21)$$

where B_h is the h th lag matrix of the polynomial $B(L) = (I - \Phi(L)L)^{-1}$.

Given the simulation draws of the data and the true SIRFs for each DGP, we assess the FAVAR estimator of the SIRFs using different measures of global activity. Specifically, we estimate three FAVARs which all have a similar specification to the above model of Stock and Watson (2016), but use different global activity measures as inputs:

1. *FAVAR-K*: the FAVAR in which the global activity measure is Kilian's index,
2. *FAVAR-BH*: the FAVAR in which the global activity measure is Baumeister and Hamilton (2019)'s OECD + 6 IP growth,
3. *FAVAR-RPC*: the FAVAR in which the global activity measure is a linear combination of the two measures and is estimated via RPC.

We then compare the SIRF estimates to the true SIRFs and assess the bias induced by using noisy measures of global activity. Regarding the lag order, we consider both fixing $p = 4$ lags and selecting the FAVAR lag length by the Akaike Information Criterion (AIC) with $p \leq 8$. Both cases delivered similar results, and in what follows we report those based on the lag order selected by AIC. For all the Monte Carlo results we use datasets with dimensions $N = 208$ and $T = 240$, and conduct 1000 repetitions.

5.3 Results

This section presents the results of the Monte Carlo analysis. For each DGP and identification approach, we focus on the corresponding average SIRFs across all Monte Carlo repetitions and assess the bias with respect to their true counterparts.

SIRFs for selected variables We first present the average SIRFs with respect to the three shocks and discuss each DGP separately. Figure 2 presents the results given the simulation draws of the

data in the first design where Kilian’s index is treated as the true global activity measure. There are three panels showing the SIRFs with respect to an oil supply shock, a global demand shock, and an oil-specific demand shock, respectively. The blue dotted line shows the average SIRFs computed using the FAVAR-K approach, along with one-standard error bands. Those computed using the FAVAR-BH and FAVAR-RPC are shown as the dashed-orange and solid-black lines, respectively. Note that here the global demand shock is always identified as the shock that raises the common component of Kilian’s index by one unit.

Across the three models, the SIRFs computed using FAVAR-RPC are visually indistinguishable from those computed based on the FAVAR-K, which is the true DGP in this analysis. In contrast, SIRFs based on the FAVAR-BH approach are clearly biased in a few cases. That said, for the case of the oil supply shock shown in Panel A, the three models produce very similar results. This is also reflected by the correlations across the three identified oil supply shocks across the three models in Table 2 which all exceed 0.95. Note that the similarity of the results is consistent with global oil production being notoriously difficult to predict (Stock and Watson, 2016).

Turning to the global demand shock in Panel B, the FAVAR-RPC estimator produces results which are quite close to the true SIRFs. This is in stark contrast to the FAVAR-BH model which generates biased SIRFs for all variables of interest in this setup. Finally, considering the oil-specific demand shock in Panel C of the table, we see that the FAVAR-BH model also gives rise to biased SIRFs in the case where the true global activity indicator is assumed to be equal to Kilian’s index. In contrast, the SIRFs using the FAVAR-RPC closely match the correct ones.

We will now compare these results with the second simulation design in which the Baumeister and Hamilton (2019) index is treated as the true global activity measure. These are provided in Figure 3. While all three models again produce similar results for the oil supply shock (Panel A), they strongly differ for the global demand shock (Panel B) and the oil-specific demand shock (Panel C). Now the SIRFs based on the FAVAR-K model are biased, but the SIRFs for the FAVAR-RPC model are essentially identical to the true SIRFs from the FAVAR-BH model. As in the previous simulation design, the bias is particularly pronounced for key variables of interest, particularly the real price of oil, real U.S. GDP and CPI inflation.

Finally, consider the case where the true global activity index is given as a combination of both measures in factor estimation. The corresponding SIRFs are presented in Figure 4. In this case, all three models produce quite similar results. The reason is that once a combination of the two measures is used for generating the simulated data, all three models explain large shares of the variance in Kilian’s index. In fact, the average R^2 of the common component of Kilian’s index over all the Monte Carlo repetitions for all three models is above 74%. Because the global demand shock is identified as an innovation to the common component of the Kilian index, the SIRFs for all three models are quite similar.

SIRFs for all variables So far we have focused on structural impulse responses for a small number of variables of interest. The advantage of FAVAR models over regular VAR analysis is that one can study the impulse responses for a large number of variables included in the model and used for factor estimation. We now assess the average of the absolute bias across all the 208 series in the dataset.

Panels A-C in Figure 5 assess the absolute bias in the FAVAR estimators of the SIRFs for an oil supply shock, a global demand shock, and an oil-specific demand shock, respectively. Using the simulation draws of the data from each DGP, the figure shows the average (across 208 series) of the absolute bias up to the three-year horizon. Specifically, using $\widehat{\text{SIRF}}_{ij}(h)$ to denote the estimated impulse response of variable i to shock j for a given FAVAR estimator, the absolute bias is constructed as

$$\sqrt{\widehat{E} \left(\widehat{\text{SIRF}}_{ij}(h) - \text{SIRF}_{ij}(h) \right)^2} \quad (22)$$

where $\widehat{E}$ denotes the average over all the Monte Carlo repetitions.

Each chart shows three lines corresponding to three FAVARs we considered. Before taking the average across different variables, we cancel the units of the SIRFs following Li et al. (2024). This is accomplished by dividing the absolute bias by $\sqrt{\frac{1}{h_{max}+1} \sum_{k=0}^{h_{max}} \text{SIRF}_{ij}^2(k)}$ (the average root mean squared value of the true SIRFs).

The figure shows that the FAVAR-RPC estimator produces substantially smaller bias in all three designs we consider. For the design combining both global activity measures (first column), the FAVAR-RPC and FAVAR-K estimators produce similar results and are preferred over the FAVAR-BH estimator. As explained above, this is consistent with the FAVAR-RPC and FAVAR-K approach explaining almost all the variance in the Kilian index when both measures are treated as noisy measures of global activity ($R^2 \geq 97\%$ on average across all Monte Carlo repetitions), while this number is only 74% for the FAVAR-BH.

In the second column of the figure using Kilian’s index as the true global activity measure, a similar result holds: the FAVAR-BH estimator now is the least preferred estimator while the FAVAR-K and FAVAR-RPC estimators produce essentially identical results. Finally, for the last design using the Baumeister and Hamilton (2019) index as the global activity measure (third column), the FAVAR-RPC and FAVAR-BH estimators now produce identical results while the FAVAR-K estimator gives rise to SIRFs with substantially larger bias.

In summary, the results in this section highlight that in the presence of noisy factor proxies, our reduced-rank PC estimator can be used to estimate structural impulse response functions which feature substantially less bias compared to approaches which assume the true factor proxies are known and observed without error.

6 Conclusion

In this paper, we have introduced a method for extending FAVARs to include multiple observable time series as noisy proxies of some true underlying factors. Our approach builds on the observation that the FAVAR model can be represented with a rank restriction when there are more observable proxies than factors. We show that the model can be estimated via an iterative procedure that alternates between principal components estimation and solving a reduced-rank regression model. We further discuss an extension adding sparsity constraints on the factor loadings for variable selection and regularization.

Our method allows researchers to include relevant information from observable proxies into the factor space without relying on the standard assumption made in FAVARs that some of the factors are perfectly measured by observable series. In a series of Monte Carlo simulations, we have documented that combining principal components with (sparse) reduced-rank regressions significantly improves factor estimation and forecast performance. It also substantially reduces the bias which arises in the presence of noisy factor proxies when estimating structural impulse response functions.

References

- Ahn, S.C., Horenstein, A.R., 2013. Eigenvalue ratio test for the number of factors. *Econometrica* 81, 1203–1227.
- Anderson, T.W., 1951. Estimating linear restrictions on regression coefficients for multivariate normal distributions. *The Annals of Mathematical Statistics* , 327–351.
- Bai, J., 2003. Inferential theory for factor models of large dimensions. *Econometrica* 71, 135–171.
- Bai, J., Ng, S., 2002. Determining the Number of Factors in Approximate Factor Models. *Econometrica* 70, 191–221.
- Bai, J., Ng, S., 2006. Confidence intervals for diffusion index forecasts and inference for factor-augmented regressions. *Econometrica* 74, 1133–1150.
- Bai, J., Ng, S., 2008. Forecasting economic time series using targeted predictors. *Journal of Econometrics* 146, 304–317.
- Bai, J., Ng, S., 2013. Principal components estimation and identification of static factors. *Journal of Econometrics* 176, 18–29.
- Bailey, N., Kapetanios, G., Pesaran, M.H., 2021. Measurement of factor strength: Theory and practice. *Journal of Applied Econometrics* 36, 587–613.
- Baumeister, C., Hamilton, J.D., 2019. Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. *American Economic Review* 109, 1873–1910.
- Bernanke, B.S., Boivin, J., Elias, P., 2005. Measuring the effects of monetary policy: a factor-augmented vector autoregressive (favar) approach. *The Quarterly Journal of Economics* 120, 387–422.
- Boivin, J., Giannoni, M.P., Mihov, I., 2009. Sticky prices and monetary policy: Evidence from disaggregated us data. *American Economic Review* 99, 350–84.
- Boivin, J., Ng, S., 2006. Are more data always better for factor analysis? *Journal of Econometrics* 132, 169–194.
- Chamberlain, G., Rothschild, M., 1983. Arbitrage, factor structure, and mean-variance analysis on large asset markets. *Econometrica* 51, 1281–1304.
- Chen, L., Huang, J.Z., 2012. Sparse reduced-rank regression for simultaneous dimension reduction and variable selection. *Journal of the American Statistical Association* 107, 1533–1545.
- Cheng, X., Liao, Z., Schorfheide, F., 2016. Shrinkage estimation of high-dimensional factor models with structural instabilities. *The Review of Economic Studies* 83, 1511–1543.

- Chudik, A., Pesaran, M.H., Tosetti, E., 2011. Weak and strong cross-section dependence and estimation of large panels. *The Econometrics Journal* 14, C45–C90.
- Friedman, J.H., Hastie, T., Tibshirani, R., 2010. Regularization paths for generalized linear models via coordinate descent. *Journal of Statistical Software* 33, 1–22.
- Giglio, S., Xiu, D., Zhang, D., 2025. Test assets and weak factors. *The Journal of Finance* 80, 259–319.
- Izenman, A.J., 1975. Reduced-rank regression for the multivariate linear model. *Journal of Multivariate Analysis* 5, 248–264.
- Kapetanios, G., Marcellino, M., 2010. Factor-GMM estimation with large sets of possibly weak instruments. *Computational Statistics Data Analysis* 54, 2655–2675. The Fifth Special Issue on Computational Econometrics.
- Kilian, L., 2009. Not all oil price shocks are alike: Disentangling demand and supply shocks in the crude oil market. *American Economic Review* 99, 1053–69.
- Kilian, L., 2019. Measuring global real economic activity: Do recent critiques hold up to scrutiny? *Economics Letters* 178, 106–110.
- Lazarus, E., Lewis, D.J., Stock, J.H., Watson, M.W., 2018. HAR inference: Recommendations for practice. *Journal of Business & Economic Statistics* 36, 541–559.
- Li, D., Plagborg-Møller, M., Wolf, C.K., 2024. Local projections vs. VARs: Lessons from thousands of DGPs. *Journal of Econometrics* 244, 105722.
- McCracken, M.W., Ng, S., 2016. FRED-MD: A monthly database for macroeconomic research. *Journal of Business & Economic Statistics* 34, 574–589.
- Moench, E., Ng, S., Potter, S., 2013. Dynamic hierarchical factor models. *The Review of Economics and Statistics* 95, 1811–1817.
- Onatski, A., 2010. Determining the number of factors from empirical distribution of eigenvalues. *The Review of Economics and Statistics* 92, 1004–1016.
- Onatski, A., 2012. Asymptotics of the principal components estimator of large factor models with weakly influential factors. *Journal of Econometrics* 168, 244–258.
- Parker, J., Sul, D., 2016. Identification of unknown common factors: Leaders and followers. *Journal of Business & Economic Statistics* 34, 227–239.
- Qian, J., Tanigawa, Y., Li, R., Tibshirani, R., Rivas, M.A., Hastie, T., 2022. Large-scale multivariate sparse regression with applications to UK Biobank. *The Annals of Applied Statistics* 16, 1891.
- Schorfheide, F., 2005. VAR forecasting under misspecification. *Journal of Econometrics* 128, 99–136.

- Stock, J.H., Watson, M.W., 2002. Forecasting using principal components from a large number of predictors. *Journal of the American Statistical Association* 97, 1167–1179.
- Stock, J.H., Watson, M.W., 2005. Implications of Dynamic Factor Models for VAR Analysis. Working Paper 11467. National Bureau of Economic Research.
- Stock, J.H., Watson, M.W., 2016. Dynamic factor models, factor-augmented vector autoregressions, and structural vector autoregressions in macroeconomics, in: *Handbook of Macroeconomics*. Elsevier. volume 2, pp. 415–525.
- Uematsu, Y., Yamagata, T., 2023a. Estimation of sparsity-induced weak factor models. *Journal of Business & Economic Statistics* 41, 213–227.
- Uematsu, Y., Yamagata, T., 2023b. Inference in sparsity-induced weak factor models. *Journal of Business & Economic Statistics* 41, 126–139.
- Velu, R., Reinsel, G.C., 1998. *Multivariate reduced-rank regression: theory and applications*. Springer New York, NY.
- Yuan, M., Lin, Y., 2006. Model selection and estimation in regression with grouped variables. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 68, 49–67.

Table 1: Monte Carlo Results I

Case	#Proxies	$R^2_{F,F}$			$S^2_{(S)RPC}/S^2_{PC}$				$\hat{r}_{ED}$	$\hat{r}_{CV}$		$\hat{q}_{CV}$	$\hat{r}_{CV}$	$\hat{q}_{CV}$	SRPC
		PC	SOFAR	RPC _(1,1)	RPC	SRPC	SOFAR	RPC _(1,1)		RPC	SRPC				
(A) Common F_2 : no restrictions imposed on F_2 loadings and $\text{var}(F_{2t}) = \text{var}(F_{1t})$															
1	1	0.93	0.93	0.86	0.93	0.93	0.93	1.64	0.95	0.95	2.03	1.90	0.10	1.90	0.10
2	5	0.93	0.94	0.92	0.92	0.92	0.89	0.95	0.90	0.93	2.04	1.44	0.57	1.42	0.60
3	10	0.93	0.94	0.93	0.93	0.93	0.92	0.89	0.90	0.93	2.04	1.32	0.68	1.30	0.73
4	20	0.93	0.94	0.94	0.94	0.93	0.96	0.78	0.81	0.83	2.04	1.15	0.86	1.12	0.90
5	20	0.93	0.94	0.92	0.92	0.92	0.97	1.02	0.95	0.99	2.04	1.45	0.56	1.43	0.61
(B) Somewhat weak F_2 : 50% of F_2 loadings set to zero and $\text{var}(F_{2t}) = \text{var}(F_{1t})$															
1	1	0.87	0.90	0.88	0.88	0.88	0.79	0.79	0.80	0.80	1.99	1.06	0.94	1.06	0.94
2	5	0.87	0.90	0.93	0.93	0.93	0.76	0.51	0.51	0.53	2.00	1.01	1.00	1.01	1.00
3	10	0.87	0.89	0.93	0.93	0.93	0.80	0.46	0.48	0.50	1.99	1.01	1.00	1.01	1.00
4	20	0.87	0.90	0.94	0.94	0.94	0.80	0.45	0.46	0.45	2.00	1.01	1.00	1.01	1.00
5	20	0.86	0.89	0.92	0.92	0.92	0.79	0.52	0.53	0.54	1.98	1.00	1.00	1.00	1.00
(C.1) Weak F_2 : 90% of F_2 loadings set to zero and $\text{var}(F_{2t}) = \text{var}(F_{1t})$															
1	1	0.51	0.55	0.88	0.69	0.69	0.80	0.33	0.54	0.54	1.26	1.00	0.50	1.00	0.50
2	5	0.51	0.56	0.91	0.62	0.69	0.85	0.26	0.64	0.52	1.30	1.00	0.31	1.00	0.49
3	10	0.51	0.57	0.88	0.56	0.68	0.83	0.32	0.68	0.52	1.33	1.00	0.19	1.01	0.48
4	20	0.51	0.56	0.80	0.51	0.66	0.82	0.54	0.76	0.57	1.28	1.00	0.06	1.00	0.41
5	20	0.51	0.55	0.78	0.51	0.63	0.85	0.57	0.80	0.61	1.28	1.00	0.06	1.00	0.38
(C.2) Weak F_2 : no restrictions imposed on factor loadings and $\text{var}(F_{2t}) = 0.1 \times \text{var}(F_{1t})$															
1	1	0.52	0.52	0.88	0.72	0.72	0.99	0.37	0.57	0.57	1.13	1.01	0.58	1.01	0.58
2	5	0.52	0.52	0.93	0.77	0.79	0.96	0.27	0.53	0.51	1.14	1.01	0.63	1.01	0.69
3	10	0.52	0.52	0.92	0.76	0.79	0.99	0.29	0.57	0.53	1.13	1.00	0.61	1.00	0.69
4	20	0.52	0.53	0.91	0.72	0.79	0.98	0.33	0.63	0.51	1.15	1.01	0.54	1.01	0.70
5	20	0.52	0.52	0.89	0.71	0.75	0.99	0.39	0.65	0.57	1.13	1.01	0.52	1.01	0.65

Note: This table reports the performance of PC, SOFAR and (S)RPC estimators detailed in Section 4.1. We consider five cases for factor proxies. In the first four cases, we start with including one proxy which is the least noisiest measure of the true factor and then included an increasing number of additional noisier proxies from one case to the next. In the fifth case, we include 20 similar proxies which are generated to be highly noisy measures of the true factor. The dataset dimensions are $N = T = 100$, and the number of simulation repetitions is 1000. The results are summarized by two statistics: $R^2_{\hat{F}^+, F}$ denotes a trace R^2 of the regression of $\hat{F}^+$ onto F ; $S^2_{(S)RPC}/S^2_{PC}$ is the ratio of the sum of the out-of-sample forecasts for all the variables and compares how close the out-of-sample forecasts based on the estimated factors are to the infeasible forecasts based on the true factors. The last five columns report the average r by estimated by the use of Onatski (2010)' ED method, and the average r and q determined by the cross-validation procedure described in Section 3.1 under both the classical RRR setting and its modification with sparsity constraints.

Table 2: Correlations among Estimated Shocks

		OS ₁	OS ₂	OS ₃	GD ₁	GD ₂	GD ₃	OD ₁	OD ₂	OD ₃
Oil supply shock (OS)	FAVAR-K (1)	1.00								
	FAVAR-BH (2)	0.95	1.00							
	FAVAR-RPC (3)	0.99	0.97	1.00						
Global demand shock (GD)	FAVAR-K (1)	0.00	0.04	0.00	1.00					
	FAVAR-BH (2)	0.03	0.00	0.02	0.37	1.00				
	FAVAR-RPC (3)	0.01	0.01	0.00	0.77	0.74	1.00			
Oil-specific demand shock (OD)	FAVAR-K (1)	0.00	0.02	0.00	0.00	0.33	-0.03	1.00		
	FAVAR-BH (2)	0.00	0.00	0.00	0.13	0.00	-0.09	0.82	1.00	
	FAVAR-RPC (3)	0.02	0.01	0.00	0.12	0.29	0.00	0.95	0.90	1.00

Note: This table reports the correlations among the shocks estimated using FAVAR-K, FAVAR-BH and FAVAR-RPC over the sample period 1985Q1 to 2014Q4. Shading denotes correlations for each one of the three shocks identified using [Kilian \(2009\)](#)'s identification scheme.

Figure 1: Graphical representation of the cross-validation procedure

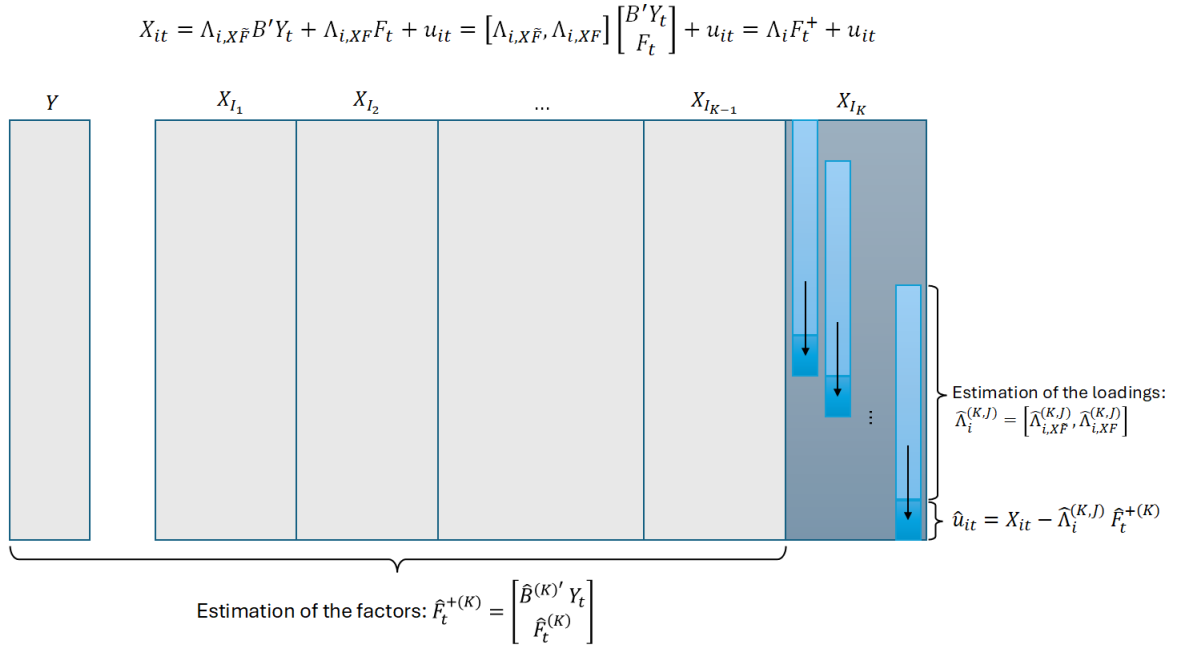
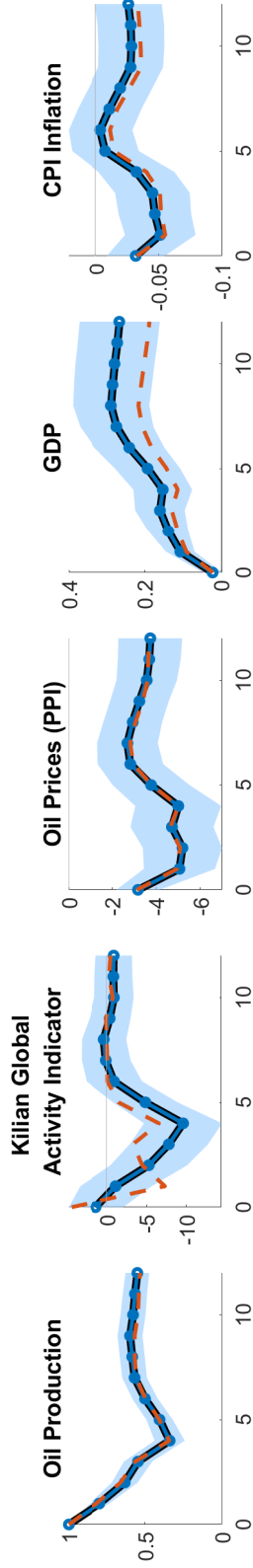
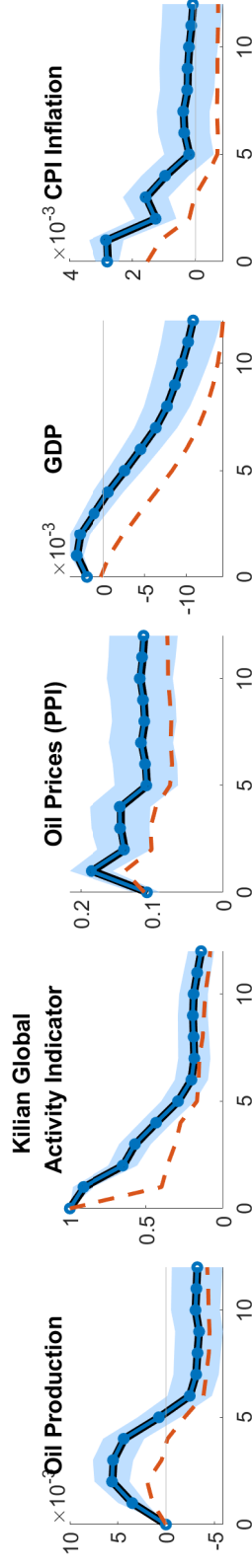


Figure 2: SIRFs with design using Kilian's global activity indicator

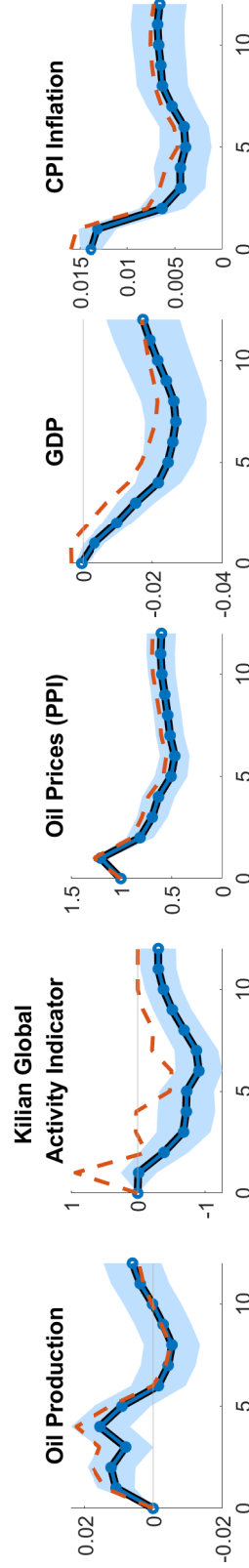
Panel A. Oil Supply Shock



Panel B. Global Demand Shock



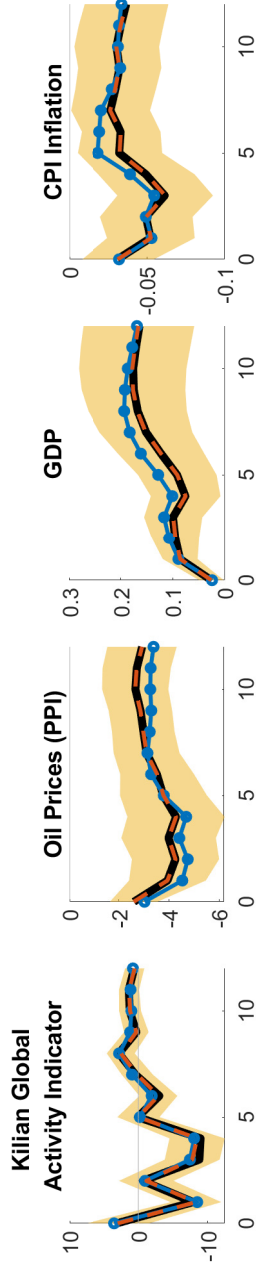
Panel C. Oil-Specific Demand Shock



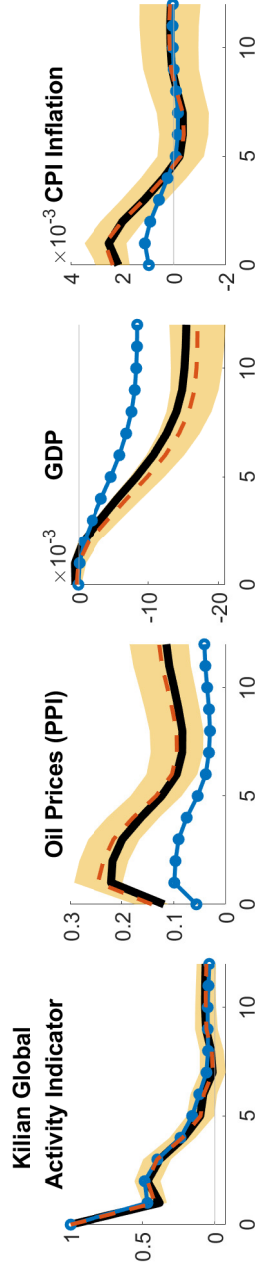
Note: This figure shows the average (over 1000 simulation repetitions) of the SIRFs from FAVAR-RPC (black solid), FAVAR-K (blue dotted ± 1 standard error bands), FAVAR-BH (orange dashed) for the oil supply shock (panel A), global demand shock (panel B) and oil-specific demand shock (panel C). Simulation draws are from the DGP in which Kilian's index is treated as the true global activity measure. The SIRFs for Kilian's global activity indicator are in standard deviation units and the responses for all other variables are in percentage points.

Figure 3: SIRFs with design using Baumeister-Hamilton OECD+6 IP growth

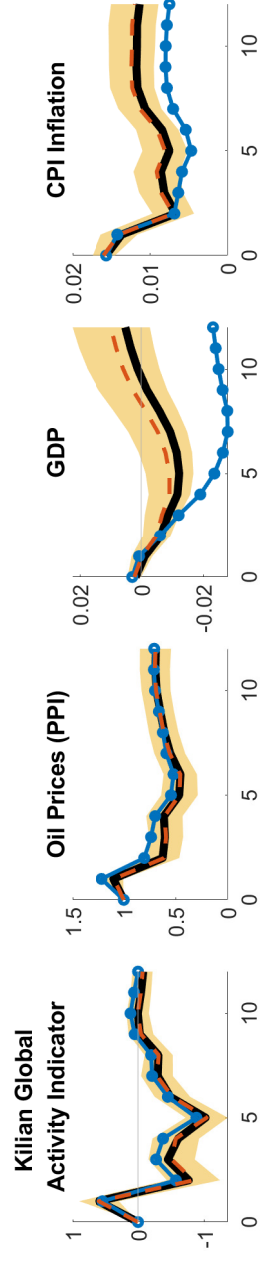
Panel A. Oil Supply Shock



Panel B. Global Demand Shock



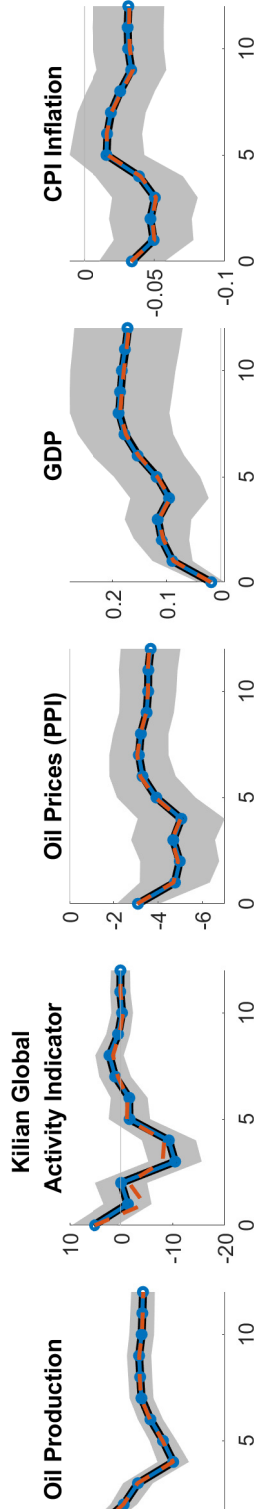
Panel C. Oil-Specific Demand Shock



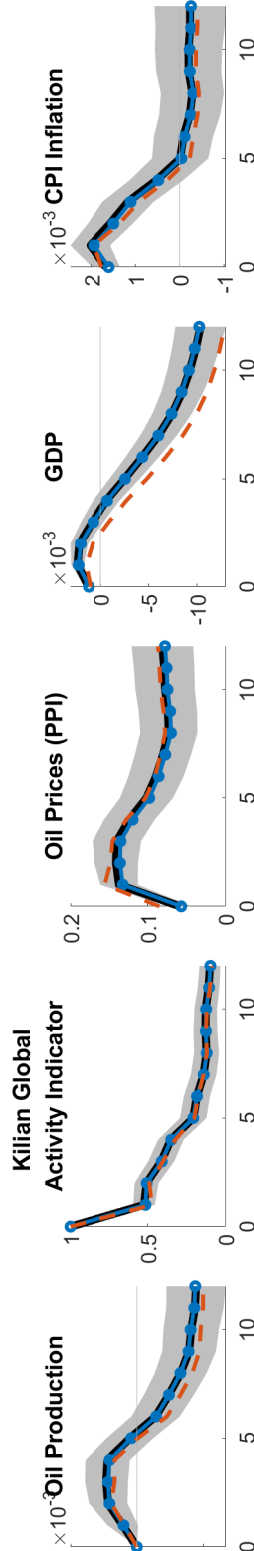
Note: This figure shows the average (over 1000 simulation repetitions) of the SIRFs from FAVAR-RPC (black solid), FAVAR-K (blue dotted), FAVAR-BH (orange dashed) ± 1 standard error bands) for the oil supply shock (panel A), global demand shock (panel B) and oil-specific demand shock (panel C). Simulation draws are from the DGP in which Baumeister and Hamilton (2019)'s OECD + 6 Emerging Market economies IP growth is treated as the true global activity measure. The SIRFs for Kilian's global activity indicator are in standard deviation units and the responses for all other variables are in percentage points.

Figure 4: SIRFs with noisy factor design

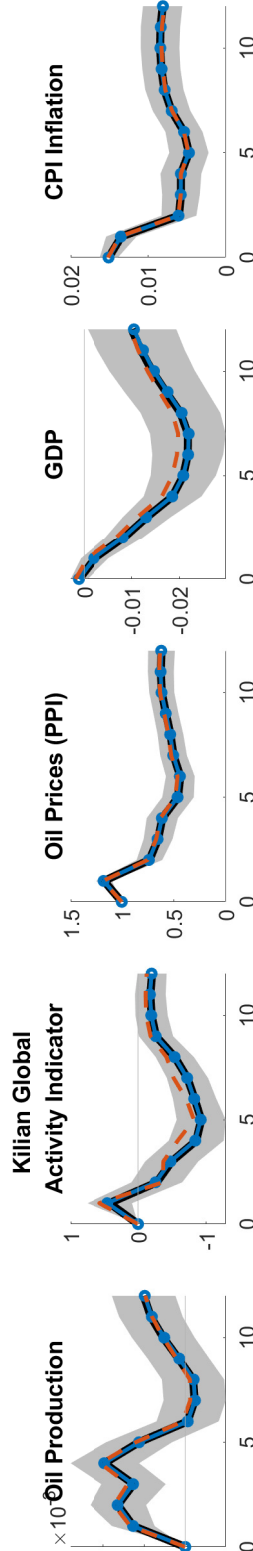
Panel A. Oil Supply Shock



Panel B. Global Demand Shock



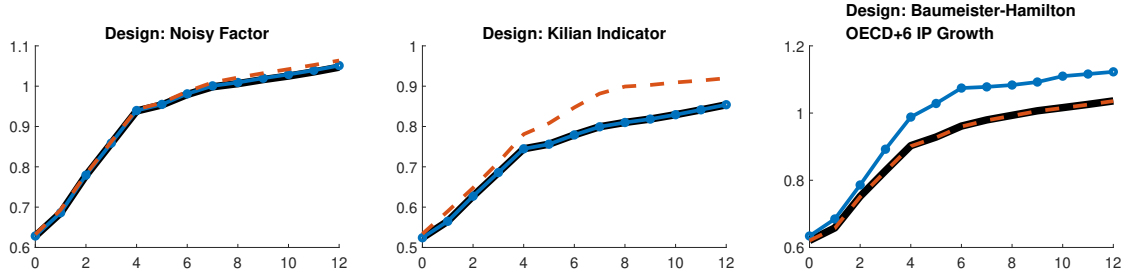
Panel C. Oil-Specific Demand Shock



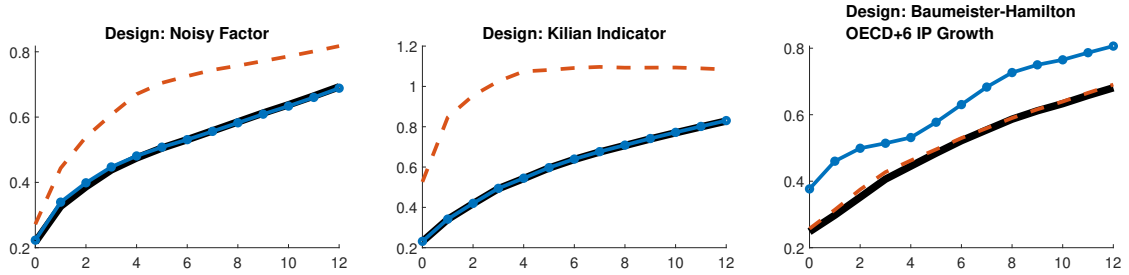
Note: This figure shows the average (over 1000 simulation repetitions) of the SIRFs from FAVAR-RPC (black solid ± 1 standard error bands), FAVAR-K (blue dotted), FAVAR-BH (orange dashed) for the oil supply shock (panel A), global demand shock (panel B) and oil-specific demand shock (panel C). Simulation draws are from the DGP which treats both Kilian's indicator and [Baumeister and Hamilton \(2019\)](#)'s OECD + 6 Emerging Market economies IP growth as two noisy measures of global activity and uses a linear combination of the two as the true global activity factor. The SIRFs for Kilian's global activity indicator are in standard deviation units and the responses for all other variables are in percentage points.

Figure 5: Average bias across 208 series

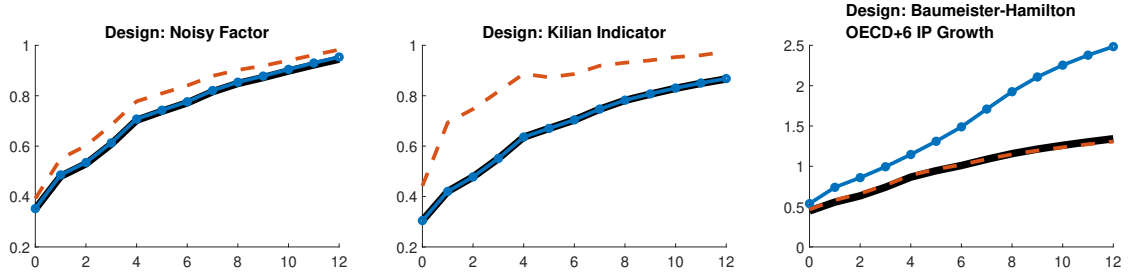
Panel A. Oil Supply Shock



Panel B. Global Demand Shock



Panel C. Oil-Specific Demand Shock



Note: This figure shows the average (over 208 variables) of the absolute bias in the FAVAR estimators of SIRFs, relative to $\sqrt{\frac{1}{13} \sum_{k=0}^{12} \text{SIRF}_{ij}^2(k)}$, for FAVAR-RPC (black solid), FAVAR-K (blue dotted) and FAVAR-BH (orange dashed). See Sections 5.2 & 5.3 for details.